

ANKAN BANSAL

Director Applied Science, Oracle, New York



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AI and Machine Learning leader with 6+ years of professional experience creating and scaling products from research to production. Directing cross-functional applied science teams to build enterprise-ready Generative AI products. Passionate about solving high-impact business problems through AI research and application.

PROFESSIONAL EXPERIENCE

Oracle Cloud Infrastructure, New York, USA

Director Applied Scientist

2025 - Present

- Heading a cross-functional team of scientists and engineers in developing and delivering a complete AI-based [Support](#) system for over 30,000 products across Oracle including a continuous improvement loop to enhance performance with usage.
- Championed and delivered cost control measures and products like automatic model routing, small language models for targeted use-cases, and user-simulations for data generation.
- Directed a diverse team of applied scientists in delivering several features like code review, NL2SQL, PL/SQL support, code provenance tracking, responsible AI, and code security agents for [Oracle Code Assist](#).

Amazon AWS, Pasadena, USA

Applied Science Manager and Senior Applied Scientist

2022 - 2025

- Managed and led a team of 14 scientists within [Amazon Quick](#) in developing large vision-language model (VLM)-based agents for enterprise users. Creating and contributing to product, science, and engineering road-maps for fast deliveries and long-term improvements.
- Spear-heading the development of evaluation approaches for automation agents. Organizing and leading deep dives into the challenges and limitations of existing approaches and devising novel approaches of evaluating the quality of agents which match human perception.
- Managed a team of 8 scientists and research engineers which built intelligent document processing (IDP) systems, called [BDA](#), within Amazon Bedrock using generative AI (VLMs and LLMs) as part of a larger org of 30 scientists and engineers. Owned the Science development for the service and collaborated with Product and Engineering owners to deploy it.
- Led the comprehensive responsible AI (RAI) assessment, and mitigation for GenAI document processing systems (BDA and Amazon Textract). BDA obtained a “low” risk rating ($\leq 1\%$ error rate) across all dimensions (race, gender, age, PII hallucination). This was instrumental in [achieving](#) the ISO/IEC 42001 accredited certification for Amazon Bedrock.
- Drove the entire data collection process for BDA by building consensus among data TPMs, SDEs and SDMs, product managers, science teams, and 150 annotators. Delivered about 15,000 documents with annotations obtained using ML-Assisted tooling developed by a multi-disciplinary team of scientists and engineers under my guidance. This tool led to over 50% higher annotation efficiency.
- Hiring, managing, mentoring, and developing scientists and engineers in the org and across the company. Serving as a “Bar Raiser” to improve hiring outcomes across the company by helping design useful loops, collecting relevant data, and driving high-quality, data-driven decisions in de-briefs.

Applied Scientist

2020 - 2022

- Created several transformer-based models for extracting structured information from invoices and receipts. Conducted pre-training, supervised fine-tuning, and evaluation of these models. These models are in production ([Amazon Textract](#)) and generating more than \$1M ARR.
- Led the science component of data generation, sourcing, and annotation for the product.

EDUCATION

Doctorate (PhD) and MS, Electrical and Computer Engineering

2020

University of Maryland, College Park, MD, USA

Advisor: Rama Chellappa

Topic: Detecting and Recognizing Humans, Objects, and their Interactions

M.Tech. and B.Tech., Electrical Engineering

2015

Indian Institute of Technology, Kanpur, India

SELECTED PUBLICATIONS AND PATENTS

SOMA-SQL: Resolving Multi-Source Ambiguity in NL-to-SQL via Synthetic Log and Execution Probing

Sai Ashish Somayajula, Marianne Menglin Liu, Chuan Lei, Fjona Parllaku, Daniel Garcia, Rongguang Wang, Syed Fahad Allam Shah, *Ankan Bansal*, Sujeeth Bharadwaj, Tao Sheng, Sujith Ravi, Dan Roth
Under Submission, 2026, [link](#)

PLSQLBench: Benchmarking LLM Systems for Executable Procedural Database Programming

Marianne Menglin Liu, Leonid Boytsov, Daniel W. Peterson, Pramuditha Perera, Rongguang Wang, Sai Ashish Somayajula, Syed Hamza Rafique, Rohit Saini, Shubham Pathak, Sujeeth Bharadwaj, Tao Sheng, Graham Horwood, Fahad Shah, *Ankan Bansal*, Sujith Ravi, Dan Roth
EMNLP 2026 Industry Track (Accepted), [link](#)

DocTr: Document Transformer for Structured Information Extraction in Documents

Haofu Liao, Aruni RoyChowdhury, Weijian Li, *Ankan Bansal*, Yuting Zhang, Zhuowen Tu, Ravi Kumar Satzoda, R. Manmatha, Vijay Mahadevan
International Conference on Computer Vision (ICCV), 2023, [link](#)

Zero-shot Object Detection

Karan Sikka, Ajay Divakaran, *Ankan Bansal*
US Patent 11,610,384, 2023, [link](#)

Object-Aware Cropping for Self-Supervised Learning

Shlok Kumar Mishra, Anshul Shah, *Ankan Bansal*, Janit K Anjaria, Abhyuday Narayan Jagannatha, Abhishek Sharma, David Jacobs, Dilip Krishnan
Conference on Lifelong Learning Agents (CoLLAs), 2023, *Journal Track*, [link](#)

Pose and Joint-Aware Action Recognition

Anshul Shah, Shlok Mishra, *Ankan Bansal*, Jun-Cheng Chen, Rama Chellappa, Abhinav Shrivastava
WACV 2022, [link](#)

Deep CNN Face Recognition: Looking at the Past and the Future

Ankan Bansal, Rajeev Ranjan, Carlos D Castillo, Rama Chellappa
Deep Learning-Based Face Analytics, 2021, [link](#)

Visual Question Answering on Image Sets

Ankan Bansal, Yuting Zhang, Rama Chellappa
European Conference on Computer Vision (ECCV), 2020, [link](#)

How are attributes expressed in face DCNNs?

Prithviraj Dhar, *Ankan Bansal*, Carlos D. Castillo, Joshua Gleason, P. Jonathon Phillips, Rama Chellappa
IEEE International Conference on Automatic Face & Gesture Recognition (FG), 2020, [link](#)

Detecting Human-Object Interactions via Functional Generalization

Ankan Bansal, Sai Saketh Rambhatla, Abhinav Shrivastava, Rama Chellappa
Thirty-Fourth AAAI Conference on Artificial Intelligence, 2020, [link](#)

Zero-Shot Object Detection

Ankan Bansal, Karan Sikka, Gaurav Sharma, Rama Chellappa, Ajay Divakaran
European Conference on Computer Vision (ECCV), 2018, [link](#)

The Do's and Don'ts for CNN-based Face Verification

Ankan Bansal, Carlos Castillo, Rajeev Ranjan, Rama Chellappa

International Conference on Computer Vision (ICCV) Workshops, 2017, [link](#)

UMDFaces: An Annotated Face Dataset for Training Deep Networks

Ankan Bansal, Anirudh Nanduri, Carlos Castillo, Rajeev Ranjan, Rama Chellappa

International Joint Conference on Biometrics (IJCB), 2017, [link](#)

SKILLS

Programming Languages and Tools: Python, PyTorch, Transformers, Tensorflow

Leadership and Management: Cross-functional teams, Communication, Strategy and Execution

SERVICE

Co-instructor for Machine Learning and Data Mining Course at Caltech (2022)

Reviewer for CV and ML conferences (CVPR, ICLR, NeurIPS etc.), and journals (IJCV, TIP, etc.)

Student Guide and Link Student, Counselling Service, IIT Kanpur, 2011-2012

OTHER INTERESTS

I am a [reader](#), a [quizzier](#), a violinist, an archer, and a chess player.